

Rajshahi University of Engineering & Technology



Department of Mechatronics Engineering

Lab Report

Course Title : Linear Integrated Circuits and Digital Systems Sessional.

Course Code : MTE 3106

Experiment No. : 03

Experiment Name : Study and Verification of Flip-Flop Circuits.

Remarks

Submitted By	Submitted To
Bayazid Habib Siddikee 3 rd year (odd Semester) Roll: 2208053 Rajshahi University of Engineering & Technology, Rajshahi	Sarafat Hussain Abhi Assistant Professor Department of MTE, Rajshahi University of Engineering & Technology, Rajshahi

Date of Submission:

Experiment No: 03

Experiment Name: Study and Verification of Flip-Flop Circuits.

Objectives

- To study the operational principles of various types of flip-flops SR, JK, Master-Slave, D, and T and verify their truth tables through practical circuit implementation.
- To analyze the characteristic equations and timing diagrams of each flip-flop type and understand how input combinations affect the output states Q and $\bar{Q}$.
- To compare the behavior of different flip-flop configurations and demonstrating their applications in sequential logic design and digital memory circuits.

Introduction

Logisim is a free, open-source digital logic circuit simulator developed by Carl Burch that allows users to design, test, and debug digital circuits on a computer using a simple drag-and-drop graphical interface. It is widely used in universities and for self-learning in digital electronics because it enables building circuits from basic gates up to complex systems such as adders, multiplexers, RAM, and even simple CPUs without any physical hardware. The software provides real-time simulation, truth tables, timing diagrams, and error highlighting, and it supports both combinational and sequential logic.

Flip-flops are fundamental bistable multivibrator circuits in digital electronics that can store one bit of binary information. Unlike combinational circuits, flip-flops are sequential circuits whose output depends not only on the present input but also on the previous state. They are the basic building blocks of registers, counters, memory units, and finite state machines. This experiment investigates five major types of flip-flops — SR, JK, Master-Slave, D, and T — each with distinct input characteristics and state-transition behavior. Understanding these devices is essential for designing stable and efficient digital systems.

Flip-Flop Types:

SR Flip-Flop/ Set-Reset Flip-Flop: The SR flip-flop is the most basic form of bistable memory element, constructed using either NOR or NAND gates. It has two inputs Set (S) and Reset (R) which control the output Q. When $S=1$ and $R=0$, the output is set to logic high; when $S=0$ and $R=1$, the output is reset to logic low. The condition $S=1, R=1$ is considered an invalid or forbidden state in NOR-based implementation, as it creates an indeterminate output. SR flip-flops serve as the foundational concept for understanding all other flip-flop types.

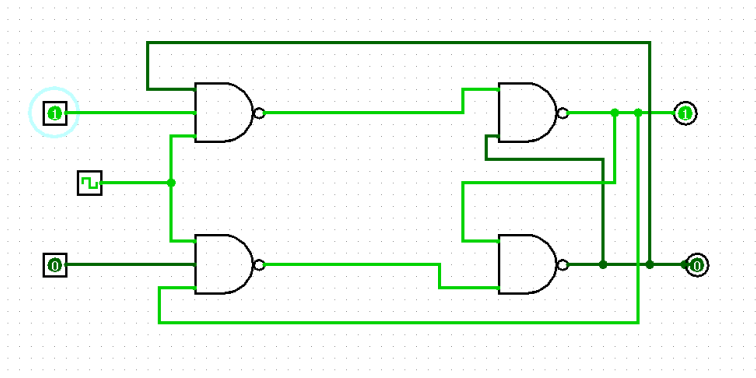


Fig 1: SR flip flop

JK Flip-Flop

The JK flip-flop is an enhanced version of the SR flip-flop that eliminates the invalid state problem. Its inputs J and K correspond to Set and Reset, respectively. When both $J=1$ and $K=1$ are applied simultaneously, the output toggles to its complement on each clock pulse, a condition not possible in the SR flip-flop. This toggle feature makes the JK flip-flop highly versatile for use in counters and frequency dividers. Its predictable behavior for all input combinations makes it one of the most widely used flip-flops in practical digital circuit design.

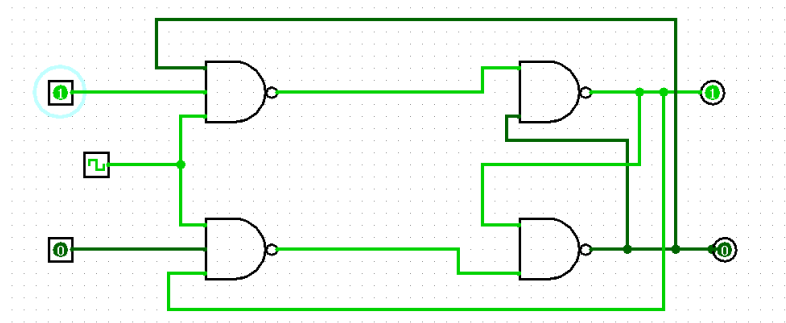


Fig 2: JK Flip Flop

Master-Slave JK Flip-Flop

The Master-Slave flip-flop is a cascade arrangement of two JK flip-flops — a master stage and a slave stage — controlled by complementary clock signals. When the clock is high, the master captures the input data; when the clock goes low, the slave transfers the master's output to the final output terminals. This two-stage mechanism effectively eliminates the racing problem or clock skew that occurs in simple edge-triggered designs. The Master-Slave configuration ensures that output changes occur only once per clock cycle, making it particularly suitable for synchronous sequential circuit applications.

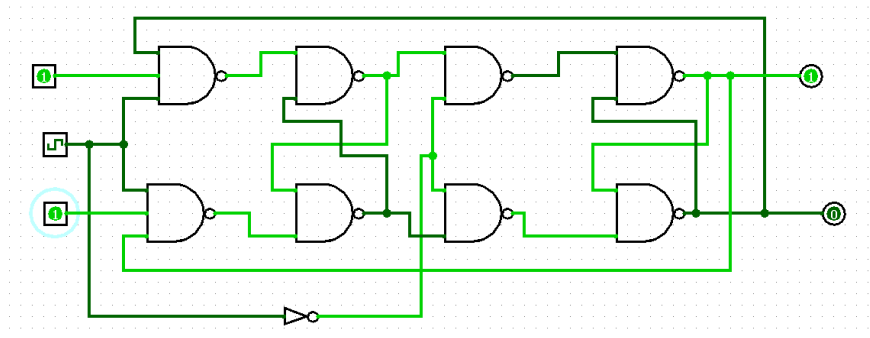


Fig 3: Master Slave JK Flip Flop

D Flip-Flop/Data Flip-Flop

The D flip-flop, also known as the Data or Delay flip-flop, has a single data input D and transfers its value to the output Q on the active edge of the clock signal. It was designed to overcome the indeterminate state problem of the SR flip-flop by ensuring the two inputs are always complementary. The output simply follows the input after one clock cycle, acting as a one-bit delay element. D flip-flops are extensively used in shift registers, data latches, pipeline

stages, and digital communication systems where data synchronization and temporary storage are required.

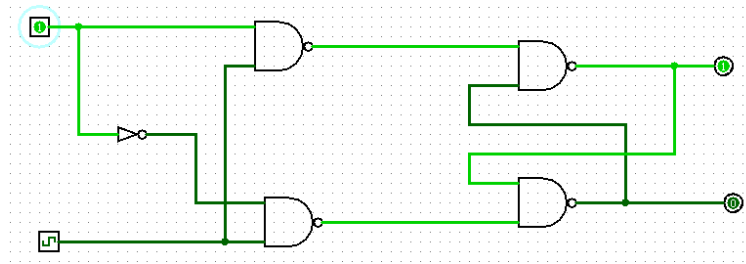


Fig 4: D flip flop

Toggle (T) Flip-Flop

The T flip-flop, or Toggle flip-flop, has a single input T that controls whether the output toggles or remains unchanged. When $T=0$, the output retains its previous state; when $T=1$, the output complements itself on each clock edge. It is essentially a JK flip-flop with both inputs tied together. This toggling property makes the T flip-flop ideal for binary counter design, frequency division, and clock pulse generation circuits. A chain of T flip-flops with $T=1$ forms a ripple counter, dividing the input frequency by successive powers of two at each stage.

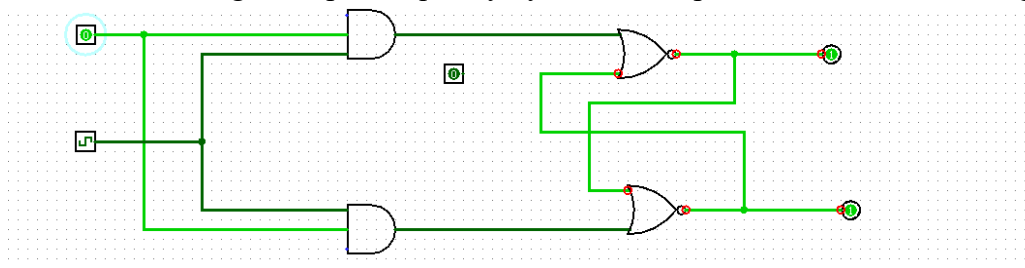


Fig 5: T flip flop

Discussion

The experiment was conducted successfully, and the truth tables of all five flip-flop types were verified. However, several difficulties were encountered during the wiring process. Due to the large number of connections involved, incorrect pin connections were made initially, which produced unexpected output values. The breadboard connections were found to be loose at multiple points, causing intermittent signal failures. Power supply fluctuations were also observed, which temporarily affected the circuit behavior. After careful rewiring and re-checking of all connections against the circuit diagrams, stable and correct outputs were eventually obtained for each flip-flop configuration.

Conclusion

This experiment successfully demonstrates the operating principles and truth table verification of SR, JK, Master-Slave, D, and T flip-flops. Each flip-flop exhibits distinct behavior based on its input configuration and clock control. These bistable circuits form the foundation of sequential digital logic and find widespread application in memory elements, counters, registers, and synchronous systems. Combinational logic handles decision-making based on current inputs, while flip-flops such as SR, JK, D, T, and Master-Slave provide the necessary memory for past states. Together, they enable the construction of everything from simple counters to full microprocessors, and Logisim makes learning and experimenting with these concepts fast, visual, and hardware-free.